

A Thermal-Electrical Co-Optimization Framework for Active Distribution Grids With Electric Vehicles and Heat Pumps

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ABSTRACT The growing electrification of transportation and heating through Electric Vehicles (EVs) and Heat Pumps (HPs) introduces both flexibility and complexity to Active Distribution Networks (ADNs). These resources provide substantial operational flexibility but also create tightly coupled thermal–electrical dynamics that challenge conventional network management. This paper proposes a unified co-optimization framework that integrates a calibrated 3R2C grey-box building thermal model into a network-constrained Optimal Power Flow (OPF). The framework jointly optimizes EVs, HPs, and photovoltaic systems while explicitly enforcing thermal comfort, Distributed Energy Resource (DER) limits, and full power flow physics. To maintain computational tractability, Second-Order Cone Programming (SOCP) relaxations are evaluated on a realistic low-voltage feeder. The analysis shows that, despite network heterogeneity violating some theoretical exactness conditions, the relaxation remains exact in practice. Comparative assessments of convex DistFlow, bus injection, and branch flow formulations reveal that convex DistFlow achieves sub-second runtimes and near-optimal performance even at high DER penetration levels. Simulations confirm the effectiveness of the proposed methodology, achieving up to a 47% reduction in daily network losses while mitigating transformer aging, eliminating voltage violations, and reducing line and transformer loading. The proposed framework consistently outperforms representative price-based and active-power-only scheduling approaches.

INDEX TERMS Active distribution networks, convex relaxation, electric vehicles, heat pumps, optimal power flow.

I. INTRODUCTION

THE ongoing transition toward electrified residential energy systems, driven by the integration of Electric Vehicles (EVs) and Heat Pumps (HPs), presents both challenges and opportunities for distribution system operators [1]. These assets offer flexibility for balancing and voltage regulation [2], but they also introduce variability and coupling across the thermal, transportation, and electrical domains. Efficient management, therefore, requires accurate modeling and scalable optimization techniques that capture the behavior of Distributed Energy Resources (DERs) and the nonlinear nature of power flows.

The literature has proposed a variety of methods to coordinate flexibility assets such as EVs, HPs, and photovoltaic (PV) systems to enhance network operation or reduce costs. In [3], a nonlinear model predictive control is applied to co-optimize residential EV and HP operation, but it lacks grid awareness and does not account for distribution network constraints. The approach in [4] addresses this by incorporating electro-thermal modeling and full network representation into a comprehensive scheduling framework; however, it relies on Particle Swarm Optimization, a heuristic method without global optimality guarantees and with high computational demands for large-scale or real-time use. Studies such as [5] and [6] highlight the potential

of residential HPs for services like frequency response and market-driven flexibility, yet they omit system-level scheduling and network-constrained optimization. Moreover, current control strategies often depend on water tank flexibility, increased self-consumption from Renewable Energy Sources (RES), or overly simplified user comfort models [7], [8], [9], but these are seldom practical due to high upfront costs, limited modeling accuracy, or their confinement to local rather than centralized schemes.

Similar developments have occurred in the domain of EVs, which are another source of demand-side flexibility. Various strategies manage EV charging through unidirectional or bidirectional flows, enabling services such as Vehicle-to-Grid [10], [11], [12], voltage regulation [13], frequency support [14], and load balancing [15]. Aggregators are frequently employed to coordinate EV fleets, streamlining their participation in energy markets [16]. Collectively, these capabilities make EVs a cost-effective tool for enhancing grid flexibility [17].

However, the effective integration of such flexible resources relies on accurate modeling of the distribution network. The power flow problem is inherently nonlinear, non-convex, and NP-hard [18], posing significant challenges for optimization due to non-global solutions, long runtimes, and possible solver failure. To address this, several convex relaxations of the Alternating Current (AC) power flow equations have been proposed [19], [20].

A. GAP ANALYSIS

Current research approaches exhibit limitations in three fundamental areas that constrain the practical implementation of flexible energy systems focusing on EVs, HPs, and PVs, which constitute the dominant and most controllable residential assets in the distribution level [16]. First, the thermal-electrical interface typically lacks adequate integration: existing studies generally do not couple HP operation with rigorous building thermodynamics, resulting in oversimplified models that cannot fully capture real-world performance dynamics. This disconnect limits the effectiveness of current optimization strategies for actual deployment scenarios.

Second, the fragmented treatment of DERs constrains the potential for system-wide optimization: most Active Distribution Network (ADN) scheduling frameworks consider only subsets of available flexibility assets, limiting the realization of multiplicative benefits from coordinated multi-asset operation and leaving substantial economic and operational value underutilized. Finally, the computational demands of existing network-aware approaches pose scalability challenges: the predominant reliance on heuristic methods and nonlinear solvers foregoes global optimality guarantees and can render them computationally intensive for real-time operation at distribution scale, thereby constraining their transition from academic studies to practical grid management tools.

B. PAPER CONTRIBUTIONS

This work addresses the identified gaps through a novel modeling and optimization framework that explicitly couples building thermal dynamics with HP electrical operation, coordinates all major distribution-level flexibility assets, and applies a convex power flow formulation. A high-fidelity calibrated 3R2C grey-box model is developed that represents building thermodynamics and maps thermal behavior to HP electrical consumption, enabling precise thermal-electrical optimization. This thermal model is embedded within a comprehensive scheduling framework that simultaneously coordinates HPs, EVs, and PVs under network constraints by relaxing the DistFlow equations to a Second-Order Cone Program (SOCP).

Therefore, the main contributions of this work are threefold:

- *Thermal-electrical coupling*: A unified framework that integrates calibrated 3R2C grey-box models directly into electrical optimization, enabling the mapping thermal comfort constraints into HP electrical flexibility.
- *Comprehensive multi-asset coordination and transformer aging assessment*: Simultaneous scheduling of HPs, EVs, and PVs within a single network-constrained Optimal Power Flow (OPF) framework, capturing realistic operating constraints and inter-asset interactions.
- *Practical convex OPF assessment*: Computational and exactness assessment of the DistFlow-SOCP formulation for realistic low-voltage distribution networks with high DER penetration levels, demonstrating its applicability for thermal-electrical co-optimization studies.

The remainder of the paper is structured as follows. Section II presents the modeling framework, including DER constraints, the 3R2C thermal model for HPs, and the DistFlow power flow formulation with its SOCP relaxation. The case study setup, including network characteristics, building thermodynamics, and DER data, is described in Section III. Section IV presents the computational comparison of OPF formulations, exactness analysis, and network operation improvements. Finally, Section V summarizes the key findings and outlines future research directions.

II. MODELING FRAMEWORK

A. NOTATIONS

Let $\mathbb{C}$ denote the set of complex numbers. Consider a directed graph $G = (\mathcal{N}, \mathcal{L})$, where the set of buses is $\mathcal{N} = \{0, 1, \dots, n\}$ with $|\mathcal{N}| = n + 1$, and the set of lines is $\mathcal{L}$, with $|\mathcal{L}| = m$. A line directed from bus i to j is denoted $i \sim j \in \mathcal{L}$ and is modeled using the series circuit with admittance: $y_{ij}^s = y_{ji}^s \in \mathbb{C}$. The complex power flow S_{ij} and current I_{ij} represent the power and current from bus i to j , $\forall i \sim j \in \mathcal{L}$. Although graph orientation is arbitrary, it is typically chosen so that bus 0 is the root node, with all lines directed outward to form a directed tree. The set of transformers is defined as $\mathcal{R}$, while DERs are defined as $\mathcal{D}$. The line impedance is $z_{ij} = z_{ji} := 1/y_{ij}^s$ where $z_{ij} = r_{ij} + \mathbf{j}x_{ij}$,

with r_{ij} and x_{ij} denoting the line resistance and reactance, respectively. The shunt conductance and susceptance at bus j are denoted by g_j and b_j , respectively. The complex power injection at bus j is $s_j := p_j + \mathbf{i}q_j$, with voltage phasor V_j . $\ell_{ij} := |I_{ij}|^2$ and $v_j := |V_j|^2$ define current and voltage magnitude square. $\bar{\bullet}$, $\underline{\bullet}$, and $\hat{\bullet}$ denote the upper bound, lower bound, and estimated value of a quantity, respectively. The set Ω_θ^D denotes the subset of Ω^D corresponding to resource type θ , for all $\theta \in \Theta$, where $\Theta = \{\text{RES}, \text{EVs}, \text{HPs}\}$ represents the distributed resource types. The set Ω^T denotes the set of discrete time steps. Finally, H denotes the time step duration.

B. DISTRIBUTED ENERGY RESOURCES

A set of electricity-consuming and producing resources is considered. Each resource is classified into a specific group within the type set $\Omega_\theta^D, \forall \theta \in \Theta$. Each resource has defined upper and lower limits for active power as described in (1). RES units have an additional constraint on their active power output due to exogenous factors such as solar irradiance and wind speed, making their maximum available power time-dependent as described in (3). Resources may operate at any level within these limits without ramp-rate constraints, as the selected time step is sufficiently large for DERs to adjust their operating points.

Reactive power control, based on the IEEE 1547–2018 standard [21], supports several local operating modes: (i) constant Power Factor (PF), (ii) voltage–reactive power (volt–var), (iii) active power–reactive power (watt–var), and (iv) constant reactive power. These modes are expressed mathematically in (4), where Q is the reactive power function and x is the state variable, e.g., for constant PF: $Q = P \cdot \tan(\cos^{-1}(PF))$, $x = P$. Alternatively, reactive power can be set via a central controller. In all cases, operation must comply with nominal limits (2). In this work, central reactive control is applied to PVs and EVs in either inductive or capacitive mode, while a constant inductive PF is assumed for HPs.

$$P_{der} \leq P_{der,t} \leq \bar{P}_{der}, \quad \forall der \in \Omega^D, t \in \Omega^T \quad (1)$$

$$Q_{der} \leq Q_{der,t} \leq \bar{Q}_{der}, \quad \forall der \in \Omega^D, t \in \Omega^T \quad (2)$$

$$P_{der,t} \leq \hat{P}_{der,t}, \quad \forall der \in \Omega_{RES}^D, t \in \Omega^T \quad (3)$$

$$Q_{der,t} = Q(x), \quad \forall der \in \Omega^D, t \in \Omega^T \quad (4)$$

C. ELECTRIC VEHICLES

For EVs, additional constraints apply to the charging process, which depends on user pReferences and connection-related information, such as arrival and departure times ($\hat{t}_{ev}^{arr}$, $\hat{t}_{ev}^{dep}$). The state-of-charge (SoC) transition function for an EV follows the equation in (5), where the energy level at time t is determined by its previous state and the charging power over the time step H . The EV's stored energy is subject to upper and lower limits, as expressed in (6), which depend on both user-defined constraints and manufacturer-imposed technical limits, with the most restrictive condition applied to ensure efficient operation. Equations (7) and (8)

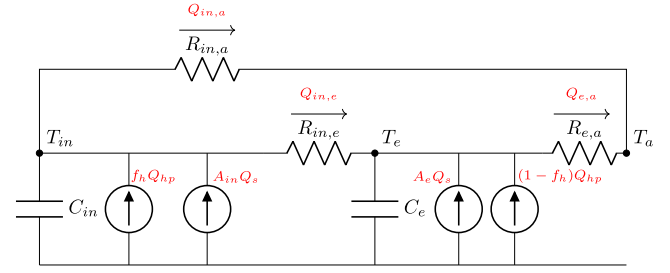


FIGURE 1. Thermal network representation of the 3R2C grey-box model.

enforce completeness by stating that there can be no active or reactive power flow when the EV is not connected. This allows certain variables to be determined directly, removing them from the optimization.

The following constraints are added $\forall ev \in \Omega_{EV}^D, t \in \Omega^T$:

$$E_{ev,t} = E_{ev,t-1} + x_{ev,t} \cdot P_{ev,t} \cdot H \quad (5)$$

$$\underline{E}_{ev} \leq E_{ev,t} \leq \bar{E}_{ev} \quad (6)$$

$$P_{ev,t}, Q_{ev,t}, x_{ev,t} = 0 \quad \text{if } t \notin [\hat{t}_{ev}^{arr}, \hat{t}_{ev}^{dep}] \quad (7)$$

$$x_{ev,t} = 1 \quad \text{if } t \in [\hat{t}_{ev}^{arr}, \hat{t}_{ev}^{dep}] \quad (8)$$

where $x_{ev,t}$ is a binary parameter that equals 1 when the EV is online, and 0 otherwise.

D. HEAT PUMPS

The operation of HPs is intrinsically linked to the thermal dynamics of the buildings they serve. To capture this bidirectional coupling, a grey-box Resistance–Capacitance (RC) thermal model is employed, based on a 3R2C configuration [22] that represents both indoor air and building envelope dynamics, as illustrated in Fig. 1 and described by (10) and (11). This model describes the evolution of indoor temperature as a function of thermal capacity, resistance parameters, and external influences such as ambient temperature and solar irradiance. In addition to modeling HP operation, the formulation includes an explicit thermal-to-electrical mapping through (9), which converts heating demand into electrical power consumption based on the operating coefficient of performance. To ensure compliance with user-defined comfort preferences, a constraint on indoor temperature is imposed, as described in (12).

The following constraints are added $\forall hp \in \Omega_{HP}^D, t \in \Omega^T$:

$$P_{hp,t} = \frac{Q_{hp,t}}{COP_{hp,t}(T_{e,t})} \quad (9)$$

$$Q_{hp,t} = \frac{T_{in,t+1} - T_{in,t}}{H} \cdot \frac{C_{in}}{f_h} - \frac{T_{e,t} - T_{in,t}}{R_{in,e} \cdot f_h} - \frac{T_{a,t} - T_{in,t}}{R_{in,a} \cdot f_h} - \frac{A_{in} \cdot Q_{s,t}}{f_h} \quad (10)$$

$$T_{e,t+1} = T_{e,t} + H \cdot \left(\frac{T_{in,t} - T_{e,t}}{C_e \cdot R_{in,e}} + \frac{T_{a,t} - T_{e,t}}{C_e \cdot R_{in,e} + \frac{(1-f_h) \cdot Q_{hp,t}}{C_e} + \frac{A_e \cdot Q_{s,t}}{C_e}} \right) \quad (11)$$

$$\underline{T}_{in,t} \leq T_{in,t} \leq \bar{T}_{in,t} \quad [^{\circ}\text{C}] \quad (12)$$

where $T_{in,t}$ [$^{\circ}\text{C}$] denotes the indoor air temperature, $T_{e,t}$ [$^{\circ}\text{C}$] is the temperature of the building envelope, and $T_{a,t}$ [$^{\circ}\text{C}$] is the ambient outdoor temperature. The term $Q_{s,t}$ [W/m^2] represents the horizontal solar irradiance, and $Q_{hp,t}$ [W] is the total heating demand, distributed between the internal and external thermal nodes according to the splitting factor $f_h \in [0, 1]$. The parameters C_{in} [J/K] and C_e [J/K] denote the thermal capacitance of the indoor air and the envelope, respectively. Thermal resistances between the indoor zone and the envelope and ambient air are given by $R_{in,e}$ [K/W] and $R_{in,a}$ [K/W], respectively, and $R_{e,a}$ [K/W] denotes the thermal resistance between the envelope and ambient air. A_{in} and A_e [m^2] define the effective surface areas for solar gains acting on the indoor and envelope nodes. The simulation timestep H [min] is used to discretize the thermal dynamics.

Model parameters for each representative building are estimated through a calibration procedure formulated as a least-squares optimization problem [23]. The objective is to identify the unknown parameter vector $\theta \in \Theta$ (e.g., thermal resistances, thermal capacitances, and solar gain coefficients) by minimizing the discrepancy between the measured indoor temperature and the temperature predicted by the grey-box model (13a) over a calibration horizon of N time steps. The constraint (13b) describes the building dynamics including those presented in (10)–(11).

$$\min_{\theta \in \Theta} \frac{1}{N+1} \sum_{k=0}^N \left(T_{in}^{\text{pred}}(k) - T_{in}^{\text{meas}}(k) \right)^2 \quad (13a)$$

$$\text{s.t.} \quad \mathbf{T}(k+1) = A_d(\theta)\mathbf{T}(k) + B_d(\theta)\mathbf{U}(k), \quad \forall k \quad (13b)$$

$$\theta_{\min} \leq \theta \leq \theta_{\max} \quad (13c)$$

where $T_{in}^{\text{pred}}(k)$ and $T_{in}^{\text{meas}}(k)$ denote the predicted and measured indoor temperatures at time step k , respectively. The vector $\mathbf{T}(k)$ contains the thermal state variables of the building, while $\mathbf{U}(k)$ represents the exogenous inputs, including outdoor temperature, solar irradiance, and heat pump thermal power. The matrices $A_d(\theta)$ and $B_d(\theta)$ are the discrete-time state-space matrices whose elements depend on the unknown physical parameters. Finally, $\theta_{\min}$ and $\theta_{\max}$ define physically meaningful parameter bounds derived from building physics considerations; see [23] for more details.

E. NETWORK AND POWER FLOW CONSTRAINTS

The Branch Flow Model (BFM) equations are adopted, with the shunt elements of the Π line model assumed to be zero, and a radial network topology considered. After an exact reformulation, voltage and current angles are eliminated from the formulation. This power flow formulation, known as the *DistFlow* model, is given in (14)–(17) and was first proposed in [24]. Voltage angles can be recovered from the original equations, but this is valid *only* for radial networks, as proven in [19]. The voltage magnitude limits in (18) ensure that the voltage at each bus i remains within safe

bounds, preserving power quality and protecting connected equipment. The ampacity limits in (19) ensure that the current magnitude on each line (i, j) does not exceed the thermal capacity $\bar{I}_{ij}$, thereby preventing overheating and damage. The line model simplification may cause discrepancies in the accuracy of $I_{ij,t}$; further details on this inconsistency are provided in [25].

The following network constraints are added $\forall t \in \Omega^T$:

$$p_j = \sum_{k:j \sim k} P_{jk} - \sum_{i:i \sim j} (P_{ij} - r_{ij}\ell_{ij}) + g_j v_j, \quad \forall j \in \mathcal{N} \quad (14)$$

$$q_j = \sum_{k:j \sim k} Q_{jk} - \sum_{i:i \sim j} (Q_{ij} - x_{ij}\ell_{ij}) + b_j v_j, \quad \forall j \in \mathcal{N} \quad (15)$$

$$v_j = v_i - 2(r_{ij}P_{ij} + x_{ij}Q_{ij}) + (r_{ij}^2 + x_{ij}^2)\ell_{ij}, \quad \forall (i, j) \in \mathcal{L} \quad (16)$$

$$\ell_{ij} = \frac{P_{ij}^2 + Q_{ij}^2}{v_i}, \quad \forall (i, j) \in \mathcal{L} \quad (17)$$

$$\underline{V}_j^2 \leq |V_{j,t}|^2 \leq \bar{V}_j^2, \quad \forall j \in \mathcal{N} \quad (18)$$

$$|I_{ij,t}|^2 \leq \bar{I}_{ij}^2, \quad \forall (i, j) \in \mathcal{L} \quad (19)$$

F. CONIC RELAXATION OF DISTFLOW

This relaxation is obtained by replacing the non-convex equality constraint (17) with a second-order cone inequality. Further details on this relaxation are provided in [19]. The proof of exactness for radial networks assumes either no upper bounds on load consumption at each bus or no lower bounds on power injection, both conditions being equivalent in this context. These assumptions generally do not hold in realistic settings, especially in systems with static loads or limited demand-response capacity. However, subsequent studies have shown that the same SOCP relaxation can remain exact under alternative sufficient conditions better aligned with practical networks. For instance, [26] presents provable conditions based on cumulative line resistance/reactance ratios and nodal voltage levels, which can be evaluated *a priori*. In particular, it provides a simple, though conservative, voltage-based criterion that guarantees exactness even in the absence of upper voltage bounds. The case study analysis will illustrate practical aspects of these conditions, and the reader is referred to [26] for a comprehensive theoretical treatment.

G. OBJECTIVE FUNCTION

The proposed AC OPF formulation schedules EVs, HPs, and PV units to minimize line losses ($C_{llosses}$) and transformer losses ($C_{tlosses}$), while also minimizing photovoltaic curtailment (C_{der}). A weighting coefficient α is applied to the curtailment cost term, set sufficiently large to prioritize curtailment reduction over power loss minimization.

The following objective function is defined $\forall t \in \Omega^T$:

$$\begin{aligned} \min_{P_{der}, Q_{der}, I, v} \quad & \alpha \cdot C_{der} + C_{llosses} + C_{tlosses} \\ \text{s.t.} \quad & \end{aligned}$$

$$\begin{aligned}
C_{der} &= \sum_{der \in \Omega_{RES}^D} (\tilde{P}_{der,t} - P_{der,t}) \\
C_{losses} &= \sum_{(i,j) \in \mathcal{L}} r_{ij} \ell_{ij,t}, \\
C_{losses} &= \sum_{(i,j) \in \mathcal{R}} P_{fe} + r_{ij} \ell_{ij,t} \quad (20)
\end{aligned}$$

III. CASE STUDY

A. TRANSFORMER AGING METRIC

The aging process of transformers is influenced by thermal effects arising from continuous loading. The transformer's lifespan is closely linked to the winding's hottest-spot temperature, which drives insulation system degradation. The IEEE C.57.91-2011 [27] standard models this relationship using an exponential expression for the aging acceleration factor. In this work, we focus on degradation effects under operating conditions where transformer loading remains below its nominal rated power. Voltage and frequency influences should also be considered when defining operational limits beyond the nameplate rating (see IEEE Std C57.12.00).

The relationship between insulation aging and temperature follows an Arrhenius-based model. The aging acceleration factor F_{AA} represents the relative insulation degradation rate compared to nominal aging at a reference hotspot temperature of 110°C , as defined in (21). A value $F_{AA} > 1$ indicates faster aging than nominal (above 110°C), while $F_{AA} < 1$ implies slower aging (below 110°C). For the scheduling horizon, the equivalent aging factor (22) is used to evaluate the impact over the entire simulation period. The θ_H variable is calculated following the method described in Section 7.2 of [27].

$$F_{AA,t} = \exp\left(\frac{15000}{383} - \frac{15000}{\theta_{H,t} + 273}\right) \quad (21)$$

$$F_{EQA} = \left(\sum_{t=1}^T F_{AA,t} \cdot H_t\right) / \sum_{t=1}^T H_t \quad (22)$$

where:

- F_{AA} : aging acceleration factor (dimensionless),
- θ_H : winding hottest-spot temperature [$^\circ\text{C}$].

B. HOUSEHOLD LOAD AND PV GENERATION PROFILES

In this study, electricity consumption data from 68 residential customers in Cyprus, recorded at 30-minute intervals over a year, are used. To focus on general consumption behavior and exclude households likely using electric heating, a seasonal filtering method compares each customer's total load consumption in winter to that in autumn. Customers with a winter-to-autumn ratio above 1.6 are excluded, being assumed to operate heating systems. After filtering, 57 customers remain, forming a refined dataset of non-heating customers. This subset is used to randomly assign Basic Load Profiles (BLPs) to feeder customers. January is

TABLE 1. 3R2C model parameters and root mean square error (RMSE) for three building types.

Parameter / Metric	SFH	OFF	TRA
$R_{in,e}[K/W]$	0.00038	0.00170	0.00028
$R_{in,a}[K/W]$	0.04996	0.03956	0.04785
$R_{e,a}[K/W]$	0.00307	0.00118	0.00153
$C_{in}[J/K]$	$4.8 \cdot 10^7$	$6.7 \cdot 10^6$	$4.7 \cdot 10^7$
$C_e[J/K]$	$8.2 \cdot 10^8$	$2.0 \cdot 10^9$	$1.4 \cdot 10^9$
$A_{in}[m^2]$	6.6	0.3	1.2
$A_e[m^2]$	7.3	41.6	19.6
f_h	0.87	0.07	0.51
Training RMSE [W]	361	61	79
Validation RMSE [W]	610	143	103
Testing RMSE [W]	527	270	151

selected for analysis as it has the highest heating demand and the most stressful winter BLP.

Statistical data are used to randomly distribute installed PV capacity across residential rooftops according to the probability mass function provided by the Cyprus Distribution System Operator (DSO). The maximum capacities for single-phase and three-phase systems are 4.16 kWp and 10.4 kWp, respectively, based on DSO limits. Due to the geographic proximity of households, the PV generation profile is assumed identical but scaled to match the installed capacity. Based on DSO data, in suburban areas, 70% of residential buildings are detached houses with a three-phase supply, and the rest have a single-phase supply. For the uncontrollable operation, a constant inductive $PF = 0.95$ is assumed.

C. EV MODELING AND DATA

The driving patterns, including departure and arrival times and trip distances, were collected through a survey targeting car owners who commute daily in Cyprus [28], [29]. The initial SoC values for EVs are randomly and uniformly assigned within predefined minimum and maximum limits of 20% and 80%, respectively. The specific EV type plays a key role in converting driving patterns into charging profiles, as it determines battery capacity and energy consumption rate. For this study, average technical specifications derived from European EV sales data are used, assuming an average consumption of 17 kWh/100km and a driving range of up to 400 km. For households with three-phase connections, 11 kW chargers are assumed, while single-phase connections use 3.7 kW chargers. By default, 50% of residential customers are assumed to own and operate EVs, unless stated otherwise. The minimum SoC is set to 30% and the maximum to 80%; for long trips, this range is extended to 20% and 90%, respectively. Further details on the dataset can be found in [30].

D. HEAT SYSTEMS AND BUILDING THERMODYNAMICS

This study assumes an Air-Source Heat Pump (ASHP) for all buildings, justified by Cyprus's mild Mediterranean

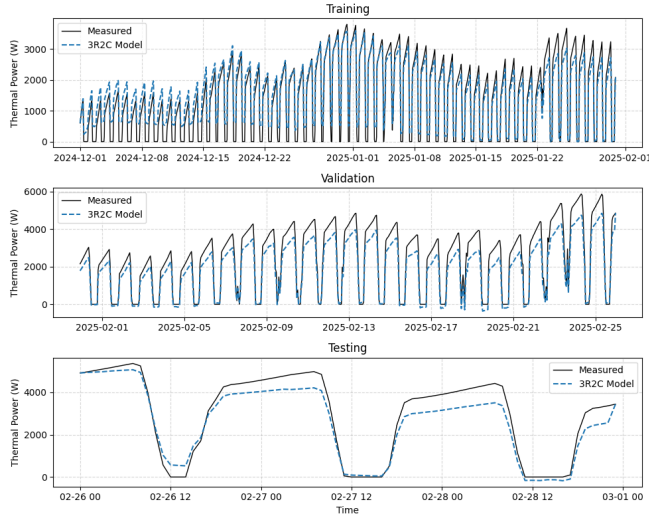


FIGURE 2. RC model tuning performance using the identified 3R2C grey-box model across three phases: training, validation, and testing.

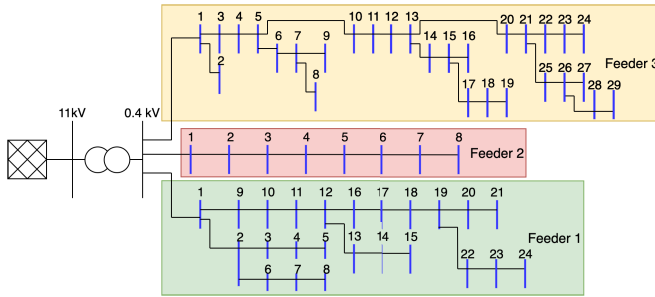


FIGURE 3. Single-line diagram of the suburban low-voltage distribution network, comprising three feeders connected to a 11/0.4 kV transformer.

climate and the lower economic viability of ground-source HPs in such conditions [31]. Three building types are considered: Single-Family House (SFH), OFFice (OFF), and Trade/Commercial Building (TRA). It is assumed that 50% of total load customers have an HP system. Of these, 60% are SFHs, 20% are commercial buildings, and the remainder are offices. The tuning procedure relies on a least-squares error metric to calibrate the model parameters; a detailed description of the tuning methodology is provided in [23]. The selected model tuning parameters are listed in Table 1, and the training, validation, and testing performance of the SFH model is shown in Fig. 2. Additional building characteristics are available in the TABULA [32] and Hotmaps [33] projects. Synthetic learning data were generated using parameter modeling proposed by the Ambience project [34].

E. DISTRIBUTION NETWORK SYSTEM MODEL

The analysis was conducted on a realistic LV distribution network provided in [35]. The network consists of a substation equipped with an 11/0.4 kV, 315 kVA delta-wye

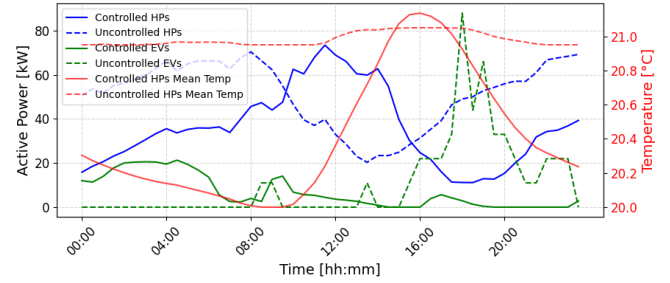


FIGURE 4. Time-series comparison of active power consumption and indoor temperature for controlled and uncontrolled operation of HPs and EVs.

(Δ -Y) transformer. Three radial feeders extend from the substation, comprising a total of 61 nodes, as depicted in Fig. 3. Each feeder supplies approximately 30 residential customers.

IV. RESULTS AND DISCUSSIONS

The SOCP optimization problem was solved using the Gurobi solver, while the IPOPT solver was adopted for the non-linear comparison problem. The implementation was carried out on a laptop equipped with an M4 Max chip, featuring a 12-core CPU and 36 GB of unified memory. The co-optimization is formulated as a multi-period OPF over 144 time steps (3 days) with a $H = 30$ min time interval. The first day served as a warm-up period, the second day as the target operation study, and the third day was included in the optimization horizon to account for next-day requirements.

A. SCHEDULE OPERATION

Figure 4 illustrates the daily power consumption and indoor temperature. Controlled HPs (solid blue line) dynamically adjust demand to follow PV production, increasing operation around noon when solar availability peaks and reducing power during evening hours (16:00–19:00). The indoor temperature remains within comfort limits by exploiting building thermal inertia. In contrast, uncontrolled HPs (dashed blue line) exhibit a relatively flat profile, operating in the opposite direction of PV output and showing limited responsiveness to network conditions.

For EVs, the controlled scenario (solid green line) distributes charging more evenly, with slight emphasis in the early morning to exploit early solar generation and prepare for departures. This strategy avoids the sharp evening peak observed in the uncontrolled case (dashed green line), where vehicles are charged immediately upon arrival, causing load synchronization and increased network stress. These results highlight the benefits of coordinated charging for both peak shaving and PV utilization.

Table 2 summarizes the average indoor temperature and total daily energy consumption of HPs and EVs under controllable and uncontrollable operation. Although the average indoor temperature differs only slightly, total energy consumption drops notably (1029 kWh vs. 1486 kWh) in the

TABLE 2. Average temperature and energy use under controllable and uncontrollable operation.

Type	Avg Temp (°C)	HP Energy (kWh)	EV Energy (kWh)
Controllable	20.4	866	163
Uncontrollable	21	1227	259

TABLE 3. Runtime [s] and objective value for OPF formulations (BIM, DistFlow, Convex DistFlow) under increasing DER penetration.

DER Penetration	BIM [36, 37]		DistFlow [24]		Convex DistFlow	
	Runtime	Obj.	Runtime	Obj.	Runtime	Obj.
0%	1.3	0.2624	0.46	0.2624	0.37	0.2624
10%	52.0	0.2402	36.5	0.2503	0.42	0.2397
20%	93.1	0.2378	66.8	0.2379	0.44	0.2328
40%	228.5	0.2355	126.5	0.2380	0.51	0.2020
60%	368.0	0.2954	174.6	0.2863	0.64	0.2465
80%	537.8	0.3630	237.2	0.3623	0.69	0.2731
100%	887.7	0.4444	367.3	0.4473	0.87	0.3089

controllable case. This demonstrates that coordinated control preserves user comfort while reducing energy use for the studied day. This reduction is not associated with a decrease in cumulative energy demand for EV users, as they distribute their charging more uniformly over time; consequently, the total energy consumption over an extended horizon remains identical to that of uncontrolled charging. In contrast, heat pumps exhibit a genuine reduction in energy consumption, as coordinated control enables a more effective exploitation of the building's thermal inertia, leading to smoother intra-day operation compared to conventional control.

Overall, intelligent scheduling preserves user comfort while improving the temporal alignment between demand and local renewable generation, thereby enhancing both energy efficiency and grid operation.

B. COMPUTATIONAL COMPLEXITY AND SCALABILITY ASSESSMENT OF OPF FORMULATIONS

To assess the computational burden and practical scalability of different OPF formulations, a benchmark analysis is conducted across three representative power flow models: the classic Bus Injection Model (BIM) [36], [37], the angle-eliminated DistFlow formulation [24] (Section E), and its convex relaxation via SOCP (Section F). The test case evaluates how runtime and solver performance vary with the penetration of EVs, HPs, and PVs in a realistic network.

The results in Table 3 reveal a clear trend in the computational burden of the three OPF formulations as the penetration of DERs (EVs, HPs, and PVs) increases. The classic BIM consistently exhibits the highest computational time, with a steep growth from 1.3s at 0% penetration to 887.7 s at 100% penetration. This growth rate indicates that BIM scales poorly under high DER penetration, making it less practical for large-scale, real-time applications. The angle-eliminated DistFlow formulation exhibits significantly

lower absolute computation times compared to the BIM; however, its scalability trend remains similar, with runtime increasing from 0.46 s to 367.3 s across the same range. The convex DistFlow formulation achieves the lowest computational times for most penetration levels, starting at 0.37 s and rising modestly to 0.87 s for 100% penetration.

Compared to BIM and DistFlow, the convex relaxation consistently yields lower objective function values, indicating an improvement in loss minimization performance. From a computational perspective, the convex DistFlow formulation delivers the most significant computational gains, maintaining sub-second runtimes even at the highest penetration levels. This combination of rapid solution time, scalability, and competitive loss minimization performance makes the convex formulation particularly well-suited for operational contexts requiring frequent OPF computations.

C. CONVEXIFICATION EXACTNESS ANALYSIS

To evaluate the exactness of the SOCP relaxation of the OPF problem for our case study, the sufficient conditions provided in Corollary 6 of [26] are considered. For convenience, these conditions are restated in Appendix.

Our analysis revealed that none of the *a priori* sufficient conditions for exactness are fully satisfied in the studied realistic network. Conditions 3, 4, 5, and 6 are violated due to line characteristics and topology. In particular, the presence of service lines with very low reactance (x) causes significant deviations in the r/x ratio across consecutive lines, making it impossible to satisfy the cumulative or neighbor-based ratio comparisons required by Conditions 3 and 4. Moreover, Condition 6 fails due to large variations in r/x ratios between neighboring lines. Similarly, Condition 3 is not satisfied because underground lines immediately after the transformer, followed by overhead lines further along the feeder, disrupt the required decreasing trend in r/x . Although Condition 2 is topologically satisfied, it prohibits any reverse power flow, a constraint invalid under operating scenarios with maximum distributed generation and minimum load. Finally, Condition 1 cannot be guaranteed in practice due to limited controllability and demand-response availability across all buses, preventing enforcement of strictly positive nominal active and reactive powers at every node.

Nevertheless, the studied case demonstrated exactness, as verified by a *post-processing* procedure. Specifically, the equality associated with the second-order cone relaxation constraint in (17) was evaluated for all time steps. Satisfying this equality confirms that the SOCP relaxation is exact. This verification procedure is equivalent to re-running an AC power flow and validating the obtained solutions at each time step, while avoiding the associated computational burden. The exactness of our case study may result from other sufficient conditions, not explicitly stated here, being satisfied, or from some of the conservative conditions actually holding under a more accurate network representation. In general, for LV networks that do not explicitly model

TABLE 4. Comparison of operational performance across the examined scheduling methods.

Method	Losses [kWh]	Min. Voltage [p.u.]	Max. Line Loading	Max. Tr. Loading	Aging Factor
Proposed	28.4	0.97	27%	46%	0.042
Uncoordinated	53.3	0.92	53%	53%	0.049
Cost-minimization [3, 39]	50.3	0.92	54%	106%	0.070
L-APC [3, 39]	35.5	0.95	29%	49%	0.051

service lines and lack capacitive reactive power control, Conditions 4 and 5 can be more easily satisfied.

Another significant point well established in the literature is that the exactness of SOCP relaxations of the DistFlow is really sensitive to the voltage-rise phenomena associated with high distributed generation and low loading levels. In such cases, the additional degrees of freedom introduced by the relaxation may allow non-physical operating points, for instance, through artificial inflation of line currents and losses, which can indirectly affect voltage profiles. This limitation has been extensively discussed in prior studies.

In the present work, the analyzed case study is characterized by a relatively high baseline demand, further increased by the electrification of heating and transportation via heat pumps and electric vehicles. Combined with the specific cable characteristics of the network, these conditions prevent voltage rise events. As a result, the operating regime considered in this study does not activate the known pathological behaviors of the SOCP relaxation. Nevertheless, readers should be aware of applications where overvoltage conditions are expected. When obtaining an exact solution is critical, iterative methodologies, some based on the same branch flow model, can be applied instead of the single-run OPF formulation proposed in this work. More specifically, in [38], the authors present a fast-solving approach in case of post-processing exactness violations.

More specifically, the relaxation gap associated with the exactness constraint (17) is evaluated using (23). If $\epsilon \leq \tau$, where τ is a predefined tolerance, the SOCP relaxation is considered exact. Otherwise, a sequential convex refinement procedure can be applied. In this case, an additional constraint in the opposite direction of the SOCP relaxation is introduced. Since this constraint is non-convex, it is linearized around the solution obtained in the previous iteration. A non-negative slack variable is further included and penalized in the objective function, progressively driving the solution toward the original branch-flow equality while preserving convexity.

$$\epsilon = \max_{(i,j) \in \mathcal{L}, t \in \mathcal{T}} (v_{i,t} l_{ij,t} - p_{ij,t}^2 - Q_{ij,t}^2). \quad (23)$$

D. DISTRIBUTION NETWORK OPERATION IMPROVEMENTS

In addition to the uncontrolled operation, the proposed methodology is compared with a *cost-minimization* scheduling approach similar to those commonly adopted in the literature, such as [3] and [39]. To enable a fair comparison,

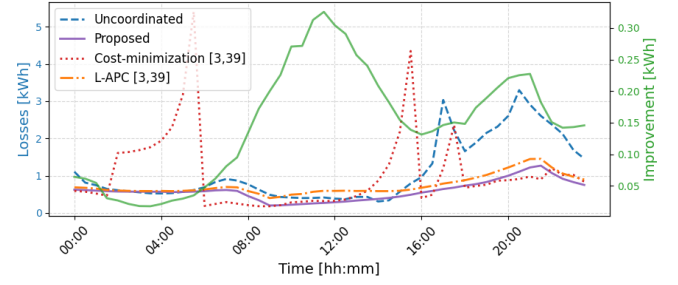


FIGURE 5. Comparison of total network losses for the proposed methodology, uncoordinated operation, cost-minimization, and L-APC. The improvement is calculated for the proposed methodology with respect to the L-APC benchmark.

these benchmark approaches were reformulated by replacing their original cost-minimization objective with the same *loss-minimization* objective adopted in this work, while retaining only active power control and neglecting network constraints (hereafter referred to as *Loss-based Active Power Control (L-APC)*). These comparisons provide further insight into the effectiveness of the proposed framework in improving network performance, reducing transformer stress, and enhancing the utilization of flexibility. The overall performance of all examined methods is summarized in Table 4.

1) LOSSES

The combined transformer and line losses over the day are shown in Fig. 5, while the overall results are summarized in Table 4. The proposed methodology reduces the daily losses from 53.3 kWh to 28.4 kWh, corresponding to a 47% reduction compared to the uncoordinated operation. The cost-minimization and L-APC benchmarks achieve daily losses of 50.3 kWh and 35.5 kWh, respectively, demonstrating that considering the electrical network and coordinated active-reactive power management provides additional loss reduction beyond active power scheduling alone. The green trace in Fig. 5 illustrates the instantaneous loss reduction of the proposed methodology with respect to the L-APC benchmark. The largest improvements are observed around midday, where coordinated reactive power support from DERs reduces feeder currents and improves voltage profiles, thereby lowering both line and transformer losses. A second improvement period occurs during the evening, when the simultaneous reconnection of EVs after arrival is coordinated across the network, preventing localized loading peaks and further reducing electrical losses.

2) VOLTAGE LEVELS

Figure 6 shows the minimum voltage trajectory throughout the day for all examined scheduling strategies. The proposed methodology consistently maintains higher minimum voltage levels by combining coordinated active and reactive power management. Reactive power support from DERs reduces feeder current and voltage drops, while the coordinated scheduling of active power alleviates network

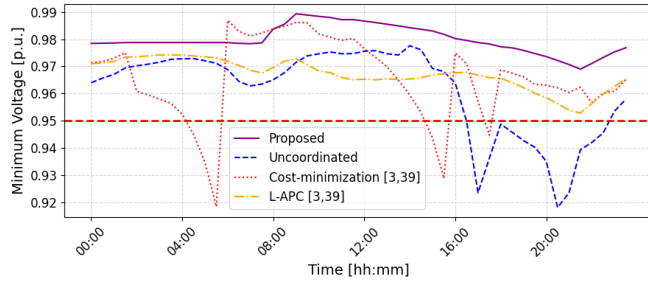


FIGURE 6. Minimum voltage profile throughout the day for the proposed methodology, uncoordinated operation, cost-minimization, and L-APC.

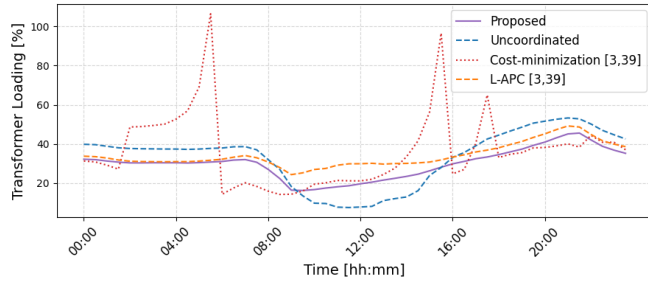


FIGURE 7. Transformer loading over the day for the proposed methodology, uncoordinated operation, cost-minimization, and L-APC.

loading during critical operating periods. The proposed methodology maintains the minimum network voltage above the 0.95 p.u. operational limit throughout the entire day, thereby eliminating voltage violations. In contrast, the uncontrolled operation experiences sustained undervoltage conditions during the evening peak due to simultaneous EV charging and HP operation. Although the cost-minimization approach partially alleviates these violations, the economically driven synchronization of flexible demand shifts the voltage stress to low-price periods, creating new short-duration voltage violations. The L-APC benchmark achieves a significant improvement over the uncontrolled case and maintains the minimum voltage close to the operational limit; however, the absence of reactive power coordination prevents it from fully exploiting the available network support.

3) TRANSFORMER LOADING AND AGING

The transformer loading profiles are presented in Fig. 7. The proposed methodology consistently maintains a smoother loading profile than the other approaches, avoiding the pronounced loading peaks observed under the cost-minimization strategy. These peaks are caused by the synchronization of flexible demand during low-price periods and exceed 100% loading for short durations. In contrast, both the proposed methodology and the L-APC benchmark distribute the demand more uniformly throughout the day, resulting in substantially lower transformer loading. A notable exception occurs around midday, where the

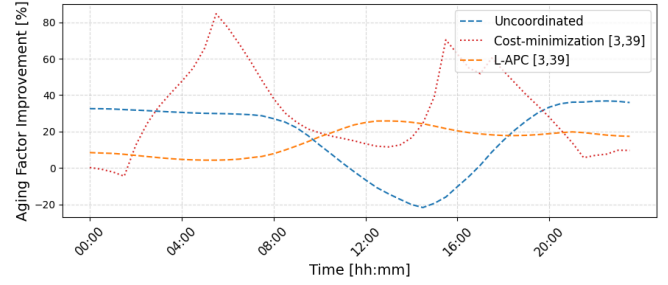


FIGURE 8. Transformer aging factor improvement over the day for the proposed methodology, uncoordinated operation, cost-minimization, and L-APC.

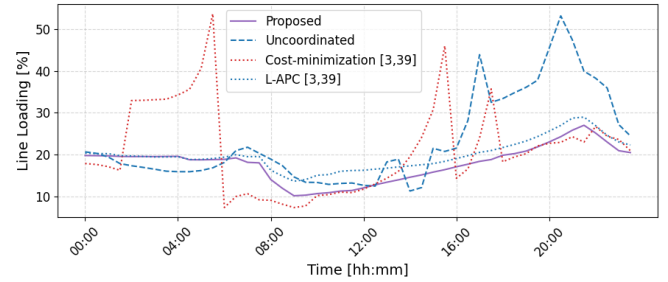


FIGURE 9. Maximum line loading comparison for the proposed methodology, uncoordinated operation, cost-minimization, and L-APC.

uncontrolled operation exhibits slightly lower transformer loading than the proposed methodology. This behavior is mainly attributed to the additional reactive power support provided by the DERs in the proposed strategy, which slightly increases the transformer apparent loading during this period.

The daily transformer aging factors are summarized in Table 4, while the corresponding transformer aging factor improvement over the day is shown in Fig. 8. The proposed methodology achieves the lowest equivalent daily aging factor among all examined strategies, corresponding to a 16% reduction relative to uncontrolled operation, a 40% reduction relative to the cost-minimization benchmark, and an 18% reduction relative to L-APC. Although the uncontrolled operation exhibits slightly lower aging than the proposed methodology during a short period around midday, this behavior is primarily attributed to the additional reactive power support provided by the DERs. The associated increase in apparent transformer loading slightly elevates the hotspot temperature during this interval. Nevertheless, this temporary increase is compensated by the significantly lower loading achieved during the remaining hours of the day, particularly during the evening EV charging period, resulting in the lowest equivalent daily transformer aging among all examined methods.

4) LINE LOADING

Fig. 9 shows the maximum line loading profiles for all examined strategies. While no thermal violations occur due

to the relatively high ampacity margins of the studied network, the proposed methodology consistently maintains the lowest line loading throughout most of the day. The L-APC benchmark follows a similar trend but exhibits slightly higher loading due to the absence of reactive power coordination. In contrast, the cost-minimization approach results in several short-duration loading spikes due to the synchronization of flexible demand during low-price periods. In more constrained distribution networks, such reductions would help prevent line overloads and improve operational reliability.

V. CONCLUSION

This paper presented a comprehensive modeling and optimization framework for the coordinated operation of EVs, HPs, and PV systems in active distribution networks. By integrating a calibrated 3R2C grey-box building thermal model into a network-constrained OPF, the proposed method establishes a direct link between thermal comfort requirements and electrical flexibility from HPs. A comparative assessment of OPF formulations demonstrated that the convexified DistFlow model offers substantial computational advantages over non-convex formulations while maintaining competitive optimality performance. The SOCP relaxation analysis under realistic LV feeder conditions revealed that theoretical exactness guarantees are often not satisfied in practice, though exactness was observed in the studied case.

Simulation results on a real Cypriot LV network demonstrated that the proposed methodology consistently outperforms all examined benchmark strategies. Compared to the uncontrolled operation, it reduces daily network losses by 47% and equivalent transformer aging by 16%, while completely eliminating voltage violations. Furthermore, compared to representative cost-minimization and active-power-only scheduling approaches, the proposed methodology achieves additional reductions of 44% and 20% in daily losses, respectively, and an 18% reduction in transformer aging relative to the active-power-only benchmark. Unlike economically driven scheduling, which tends to synchronize flexible demand during low-price periods, the proposed network-aware coordination avoids undesirable loading peaks, maintains improved voltage profiles, and reduces both transformer and line loading throughout the day. These findings demonstrate that integrating network constraints together with coordinated active and reactive power management significantly enhances both operational reliability and energy efficiency in future electrified systems.

Future work will also explore the integration of Model Predictive Control (MPC) strategies to enable real-time, receding-horizon optimization of EV and HP operation. This extension will allow the framework to adapt dynamically to forecast deviations, enhance robustness under uncertainty, and bridge the gap between day-ahead scheduling and real-time control.

APPENDIX

For reference, this appendix summarizes the sufficient conditions derived in Corollary 6 of [26], which establish when the SOCP relaxation of the OPF problem is exact. These conditions ensure that, if any of them is satisfied, the solution obtained from the relaxed formulation coincides with the optimal solution of the original non-convex OPF. The conditions are listed below:

- (1) For each $j \in \mathcal{N} \setminus \{0\}$, $\bar{p}_j \leq 0$, $\bar{q}_j \leq 0$.
- (2) For all $(i, j) \in \mathcal{L}$, $\frac{r_{i,j}}{x_{i,j}} \geq \frac{R_i}{X_i}$, with $\bar{p}_j \leq 0$.
- (3) For all $(i, j), (j, k) \in \mathcal{L}$, $\frac{r_{i,j}}{x_{i,j}} \geq \frac{r_{j,k}}{x_{j,k}}$, with $\bar{p}_j \leq 0$.
- (4) For all $(i, j) \in \mathcal{L}$, $\frac{x_{i,j}}{r_{i,j}} \geq \frac{X_i}{R_i}$, with $\bar{q}_j \leq 0$.
- (5) For all $(i, j), (j, k) \in \mathcal{L}$, $\frac{x_{j,k}}{r_{j,k}} \geq \frac{x_{i,j}}{r_{i,j}}$, with $\bar{q}_j \leq 0$.
- (6) For all $(i, j), (j, k) \in \mathcal{L}$, $\frac{r_{j,k}}{x_{j,k}} = \frac{r_{i,j}}{x_{i,j}}$.

where R_i and X_i define the cumulative resistance and reactance from the root node to bus i .

REFERENCES

- [1] A. Navarro-Espinosa and L. F. Ochoa, "Probabilistic impact assessment of low carbon technologies in LV distribution systems," *IEEE Trans. Power Syst.*, vol. 31, no. 3, pp. 2192–2203, May 2016.
- [2] T. Masuta and A. Yokoyama, "Supplementary load frequency control by use of a number of both electric vehicles and heat pump water heaters," *IEEE Trans. Smart Grid*, vol. 3, no. 3, pp. 1253–1262, Sep. 2012.
- [3] A. Petrucci, C. Vallianos, A. Buonomano, B. Delcroix, and A. Athienitis, "Coordinated load management of building clusters and electric vehicles charging: An economic model predictive control investigation in demand response," *Energy Convers. Manage.*, vol. 339, Sep. 2025, Art. no. 119965.
- [4] J. Hao et al., "Optimal scheduling of active distribution network considering symmetric heat and power source-load spatial-temporal characteristics," *Appl. Energy*, vol. 373, Nov. 2024, Art. no. 123974.
- [5] M. T. Muhssin, L. M. Cipcigan, N. Jenkins, S. Slater, M. Cheng, and Z. A. Obaid, "Dynamic frequency response from controlled domestic heat pumps," *IEEE Trans. Power Syst.*, vol. 33, no. 5, pp. 4948–4957, Sep. 2018.
- [6] E. Georges, B. Cornélusse, D. Ernst, V. Lemort, and S. Mathieu, "Residential heat pump as flexible load for direct control service with parametrized duration and rebound effect," *Appl. Energy*, vol. 187, pp. 140–153, Feb. 2017.
- [7] B. P. Bhattarai, B. Bak-Jensen, J. R. Pillai, and M. Maier, "Demand flexibility from residential heat pump," in *Proc. IEEE PES Gen. Meeting Conf. Expo.*, Jul. 2014, pp. 1–5.
- [8] T. Kemmler and B. Thomas, "Design of heat-pump systems for single- and multi-family houses using a heuristic scheduling for the optimization of PV self-consumption," *Energies*, vol. 13, no. 5, p. 1118, Mar. 2020.
- [9] T. Beck, H. Kondziella, G. Huard, and T. Bruckner, "Optimal operation, configuration and sizing of generation and storage technologies for residential heat pump systems in the spotlight of self-consumption of photovoltaic electricity," *Appl. Energy*, vol. 188, pp. 604–619, Feb. 2017.
- [10] K. M. Tan, V. K. Ramachandramurthy, and J. Y. Yong, "Integration of electric vehicles in smart grid: A review on vehicle to grid technologies and optimization techniques," *Renew. Sustain. Energy Rev.*, vol. 53, pp. 720–732, Jan. 2016.
- [11] M. Falahi, H.-M. Chou, M. Ehsani, L. Xie, and K. L. Butler-Purry, "Potential power Quality benefits of electric vehicles," *IEEE Trans. Sustain. Energy*, vol. 4, no. 4, pp. 1016–1023, Oct. 2013.
- [12] L. Carradore and R. Turri, "Electric vehicles participation in distribution network voltage regulation," in *Proc. 45th Int. Univ. Power Eng. Conf.*, 2010, pp. 1–6.
- [13] A. Kotsonias, L. Hadjidemetriou, M. Asprou, and C. G. Panayiotou, "Operational challenges and solution approaches for low voltage distribution grids—A review," *Electric Power Syst. Res.*, vol. 239, Feb. 2025, Art. no. 111258.
- [14] P. Kushwaha, V. Prakash, S. Yamujala, and R. Bhakar, "Fast frequency response constrained electric vehicle scheduling for low inertia power systems," *J. Energy Storage*, vol. 62, Jun. 2023, Art. no. 106944.

- [15] S. Weckx and J. Driesen, "Load balancing with EV chargers and PV inverters in unbalanced distribution grids," *IEEE Trans. Sustain. Energy*, vol. 6, no. 2, pp. 635–643, Apr. 2015.
- [16] F. Gonzalez Venegas, M. Petit, and Y. Perez, "Active integration of electric vehicles into distribution grids: Barriers and frameworks for flexibility services," *Renew. Sustain. Energy Rev.*, vol. 145, Jul. 2021, Art. no. 111060.
- [17] S. Habib, M. M. Khan, F. Abbas, and H. Tang, "Assessment of electric vehicles concerning impacts, charging infrastructure with unidirectional and bidirectional chargers, and power flow comparisons," *Int. J. Energy Res.*, vol. 42, no. 11, pp. 3416–3441, Sep. 2018.
- [18] D. Bienstock and A. Verma, "Strong NP-hardness of AC power flows feasibility," *Operations Res. Lett.*, vol. 47, no. 6, pp. 494–501, Nov. 2019.
- [19] M. Farivar and S. H. Low, "Branch flow model: Relaxations and convexification—Part II," *IEEE Trans. Power Syst.*, vol. 28, no. 3, pp. 2565–2572, Aug. 2013.
- [20] B. Kocuk, S. S. Dey, and X. A. Sun, "Strong SOCP relaxations for the optimal power flow problem," *Oper. Res.*, vol. 64, no. 6, pp. 1177–1196, Dec. 2016.
- [21] Standard for Interconnection and Interoperability of Distributed Energy Resources With Associated Electric Power Systems Interfaces, IEEE Standard 1547-2018, 2018.
- [22] H. Harb, N. Boyanov, L. Hernandez, R. Streblov, and D. Müller, "Development and validation of grey-box models for forecasting the thermal response of occupied buildings," *Energy Buildings*, vol. 117, pp. 199–207, Apr. 2016.
- [23] S. Panagi, L. Aresti, P. Christodoulides, C. Spanias, and P. Aristidou, "Tuning performance of grey-box models for thermal building applications," *Smart Energy*, vol. 22, May 2026, Art. no. 100239.
- [24] M. Baran and F. F. Wu, "Optimal sizing of capacitors placed on a radial distribution system," *IEEE Trans. Power Del.*, vol. PD-4, no. 1, pp. 735–743, Jan. 1989.
- [25] K. Christakou, D.-C. Tomozei, J.-Y. Le Boudec, and M. Paolone, "AC OPF in radial distribution networks—Part I: On the limits of the branch flow convexification and the alternating direction method of multipliers," *Electr. Power Syst. Res.*, vol. 143, pp. 438–450, Feb. 2017.
- [26] N. Li, L. Chen, and S. H. Low, "Exact convex relaxation of OPF for radial networks using branch flow model," in *Proc. IEEE 3rd Int. Conf. Smart Grid Commun. (SmartGridComm)*, Nov. 2012, pp. 7–12.
- [27] Guide for Loading Mineral-Oil-Immersed Transformers and Step-Voltage Regulators, IEEE Standard C57.91-2011, 2012, pp. 1–123.
- [28] P. Sokrati and P. Charalampou, "The impacts of EVs in Cyprus distribution system," Ph.D. thesis, CYPRUS Univ. Technol., Sep. 2021.
- [29] S. Panagi, P. Therapontos, C. Spanias, and P. Aristidou, "Optimal operation of electric vehicles to enhance the flexibility provision in active distribution grids," in *Proc. IEEE Kiel PowerTech*, Jun. 2025, pp. 1–6.
- [30] S. Panagi, P. Therapontos, C. Spanias, and P. Aristidou, "Impact of uncontrolled electric vehicle charging on unbalanced suburban low-voltage networks," in *Proc. IEEE PES Innov. Smart Grid Technol. Conf. Eur. (ISGT Europe)*, Oct. 2025, pp. 1–5.
- [31] A. S. Gaur, D. Z. Fitiwi, and J. Curtis, "Heat pumps and our low-carbon future: A comprehensive review," *Energy Res. Social Sci.*, vol. 71, Jan. 2021, Art. no. 101764.
- [32] TABULA Project Team.(2012). *TABULA: Typology Approach for Building Stock Energy Assessment*. [Online]. Available: <https://episcopes.eu/welcome/>
- [33] (2022). Hotmaps Project: Heating and Cooling Open Source Tool. Accessed: Dec. 8, 2025. [Online]. Available: <https://www.hotmaps-project.eu/>
- [34] I. Jankovic, X. Fernandez, and J. Diriken, "Database of grey-box model parameter values for EU building typologies," *Ambience Project*, Dec. 2021.
- [35] S. Panagi, "SPS-L/Cyprus-Grids: Complete network—Electricity authority of Cyprus," *Tech. Rep.*, Oct. 2025, doi: [10.5281/zenodo.17369365](https://doi.org/10.5281/zenodo.17369365).
- [36] F. D. Kanellos, "Optimal scheduling and real-time operation of distribution networks with high penetration of plug-in electric vehicles," *IEEE Syst. J.*, vol. 15, no. 3, pp. 3938–3947, Sep. 2021.
- [37] J. Peschon, D. W. Bree, and L. P. Hajdu, "Optimal power-flow solutions for power system planning," *Proc. IEEE*, vol. 60, no. 1, pp. 64–70, 1972.
- [38] P. Andrianesis, M. Caramanis, and N. Li, "Optimal distributed energy resource coordination: A decomposition method based on distribution locational marginal costs," *IEEE Trans. Smart Grid*, vol. 13, no. 2, pp. 1200–1212, Mar. 2022.
- [39] N. Stylianides, L. Tzirovani, S. Timotheou, and L. Hadjidemetriou, "A heat pump optimization-based scheduling to minimize energy cost and ensure thermal comfort," in *Proc. IEEE PES Innov. Smart Grid Technol. Conf. Eur. (ISGT Europe)*, Oct. 2025, pp. 1–5.



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